Augmentation and leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Check the target. Track the parent. Split the group. Match the budget.

Audit augmentation with complete label rules,lineage,pixel overlap and exposure counts.

Keep one habit: Visible row/column transformations → tie rule → pictured group boundary → complete P–T registry and faulty manifests → matched-budget arithmetic → independent new split.

MAIN LESSON · SECTION 1

Did the transformation preserve the label—and the split boundary?

> Track target meaning,source families,exact duplicates and the actual training budget.

Learning goals

I can check whether a transformation preserves the task label.

I can keep related examples in one split using **source group**s.

I can compare **augmentation** fairly without inflating independent **evidence**.

Retrieve the narrow image-model evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why must a checkpoint be chosen before inspecting final evaluation results?

> **My starting explanation**

End goal

By the end, I can explain **augmentation** and **leakage** using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Check whether a transformation preserves the task label

Keep related examples in one split using **source group**s

Compare **augmentation** fairly without inflating independent **evidence**

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Judge augmentation by the declared target rule

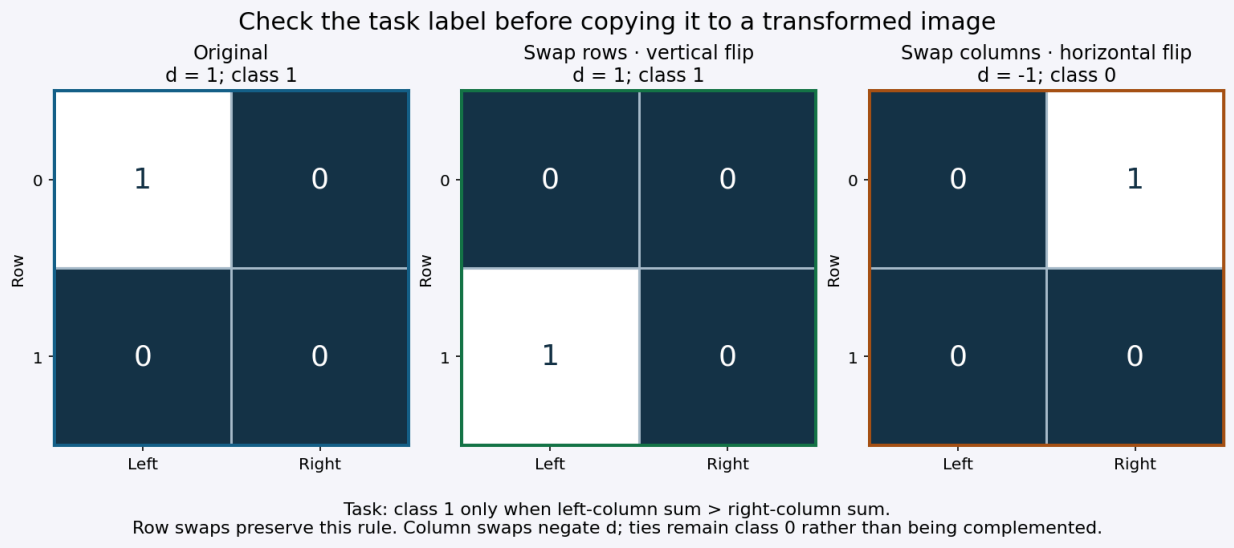
> For this arithmetic task,row and column swaps have different consequences.

Complete target and three transformed cases

Return to the binary image task: class 1 means the left column sum is strictly larger than the right column sum; equal sums belong to class 0. Let d be left sum minus right sum.

An **augmentation** is useful only under a stated task and transformation rule. More image files do not automatically mean more independent **evidence**.

Operation on rows [1, 0] and [0, 0]	Result rows	d	Class
Original	[1, 0] and [0, 0]	1	1
Swap top and bottom rows	[0, 0] and [1, 0]	1	1
Swap left and right columns	[0, 1] and [0, 0]	-1	0



Prove the label rule and name its scope

Swapping rows preserves each column sum, so it preserves this task label for every valid image. Swapping columns changes d to -d. For nonzero d the class changes; for d = 0 it stays class 0.

Blindly copying the original label after every flip would create incorrect **training** targets.

A transformation may be valid for one task and invalid for another. A crop can remove the **evidence** needed for a label.

Choose allowed transformations using the task definition and **training evidence**, document the assumptions, and evaluate the choice on validation data.

Name given and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace the target before creating a derived training record

> Keep the original image,parent,transformation and justified target together.

Function assumptions and justified transformed labels

This code assumes valid 2 by 2 binary lists. It constructs new lists instead of changing the **input**. The label function encodes the classroom task rule, not a general method for recovering unknown real-world labels.

If an allowed transformation changes a known target, the transformed example needs a justified new label. Do not invent labels for ambiguous cases.

Keep the source ID, transformation and resulting label with every derived image so a teacher can audit what happened.

Complete positive,negative and equality cases

Complete known-rule case	Original d / class	After row swap	After column swap
[[1, 0], [0, 0]]	1 /1	1 /1	–1 /0
[[0, 1], [0, 0]]	–1 /0	–1 /0	1 /1
Tie[[1, 0], [0, 1]]	0 /0	0 /0	0 /0; not complemented

Known-rule proof and experiment limits

The label function is the supplied mathematical task, not a tool for inferring unknown real-world labels. Row swapping preserves each column sum. Column swapping negates d, so it changes the binary class only when d is nonzero. The original functions create new lists and leave their input unchanged.

The complete audit program later checks all 16 binary2×2 patterns for this rule and builds an explicitly fictional provenance registry. No augmented model is trained or improved score claimed by these audits.

Runnable nonmutating row/column transformations

```
def difference(x):
    return x[0][0] + x[1][0] - x[0][1] - x[1][1]

def label(x):
    return int(difference(x) > 0)

def swap_rows(x):
    return [x[1][:], x[0][:]]

def swap_columns(x):
    return [row[::-1] for row in x]

x = [[1, 0], [0, 0]]
assert label(x) == label(swap_rows(x)) == 1
assert label(swap_columns(x)) == 0
tie = [[1, 0], [0, 1]]
assert label(tie) == label(swap_columns(tie)) == 0
assert x == [[1, 0], [0, 0]]
```

Connect the visual change to its target rule

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

Keep a transformed record with its source family

> A new filename does not make a new independent source.

Current equality-boundary task

- For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured cross-split derived record

- Change: Move D's flipped copy into **training**.

▼ Check after predicting

Prediction: Final-test group separation fails.

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Count records and source identities separately

> A paper provenance count does not approve a transformation for every task.

Complete pictured source facts and conditional label scope

Keep every augmented image with its original **source group**.

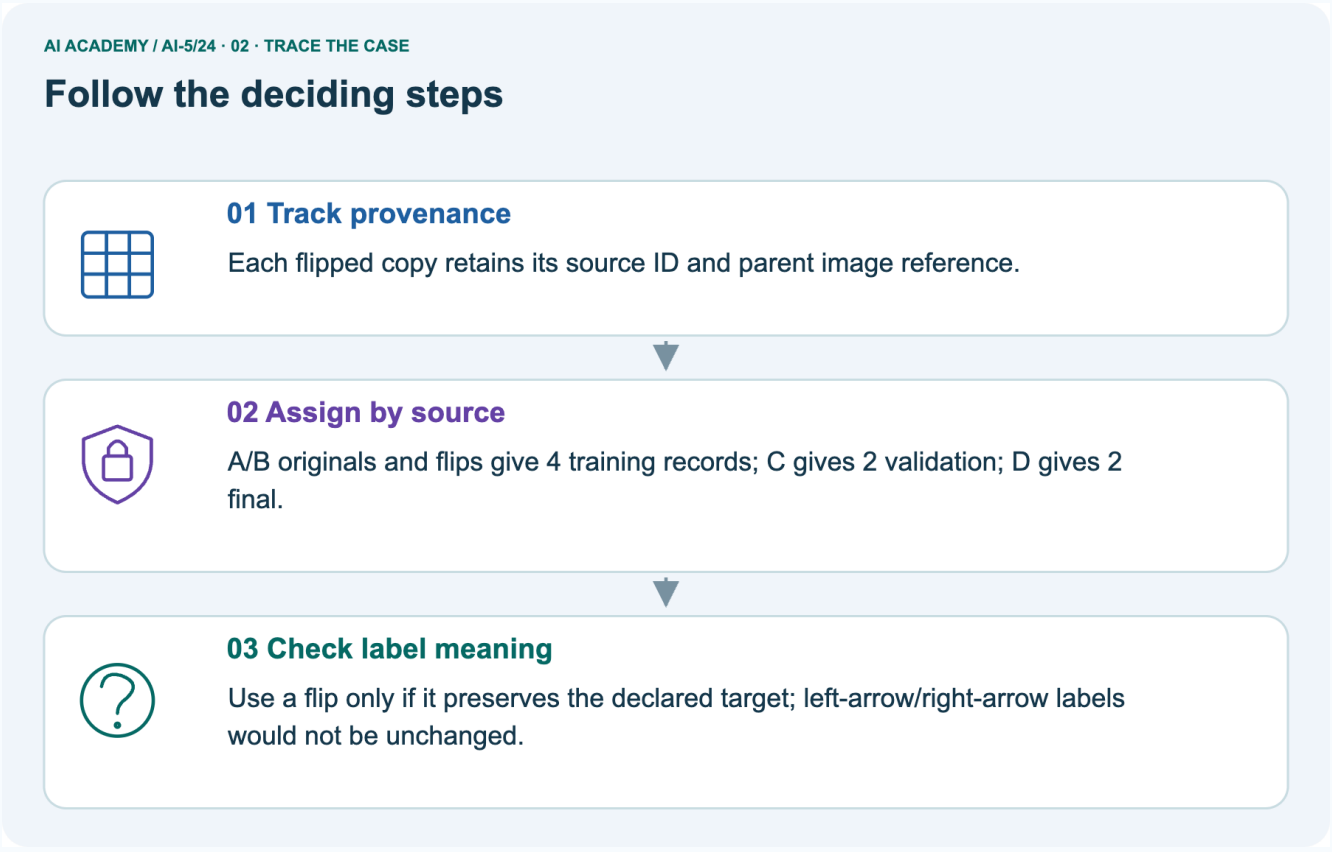
Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate provenance/counting example with local source names A–D, not the six image IDs from Week23. Each source has an original plus a horizontal-flip record. A/Btraining gives 4 records from 2 groups; Cvalidation and Dfinal each have 2 records from 1 group.

These counts do not certify label validity:for our left-versus-right target, a horizontal flip generally changes the class and needs its justified target. Moving D’s derived record to training crosses the final source boundary. The new P–T registry below uses verified row swaps and retains only original evaluation records.

Trace parent lineage and group assignment



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Track provenance	Each flipped copy retains its source ID and parent image reference.
2. Assign by source	A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.
3. Check label meaning	Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

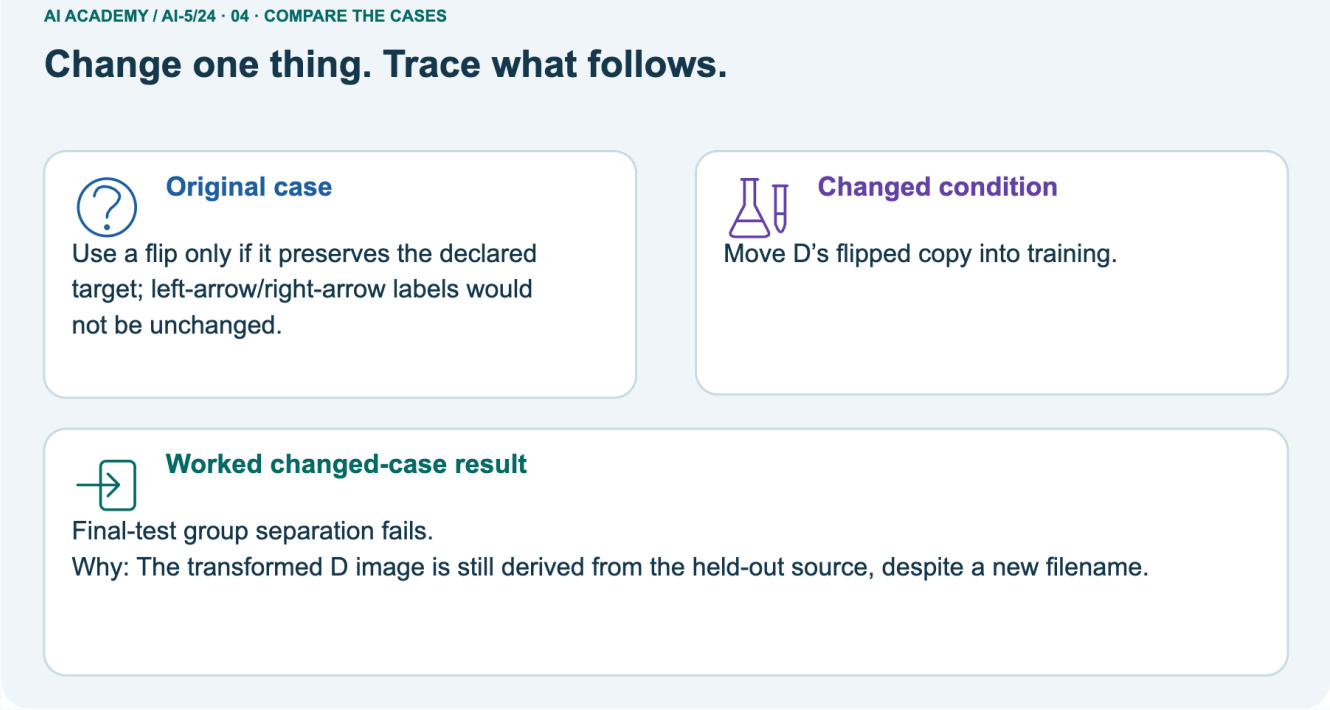
An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Predict the moved D derived record

- Change one thing: Move D's flipped copy into **training**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain independent group count and sampling limits

- **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

- Read the labelled picture: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the earlier A/A2filename trap

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.

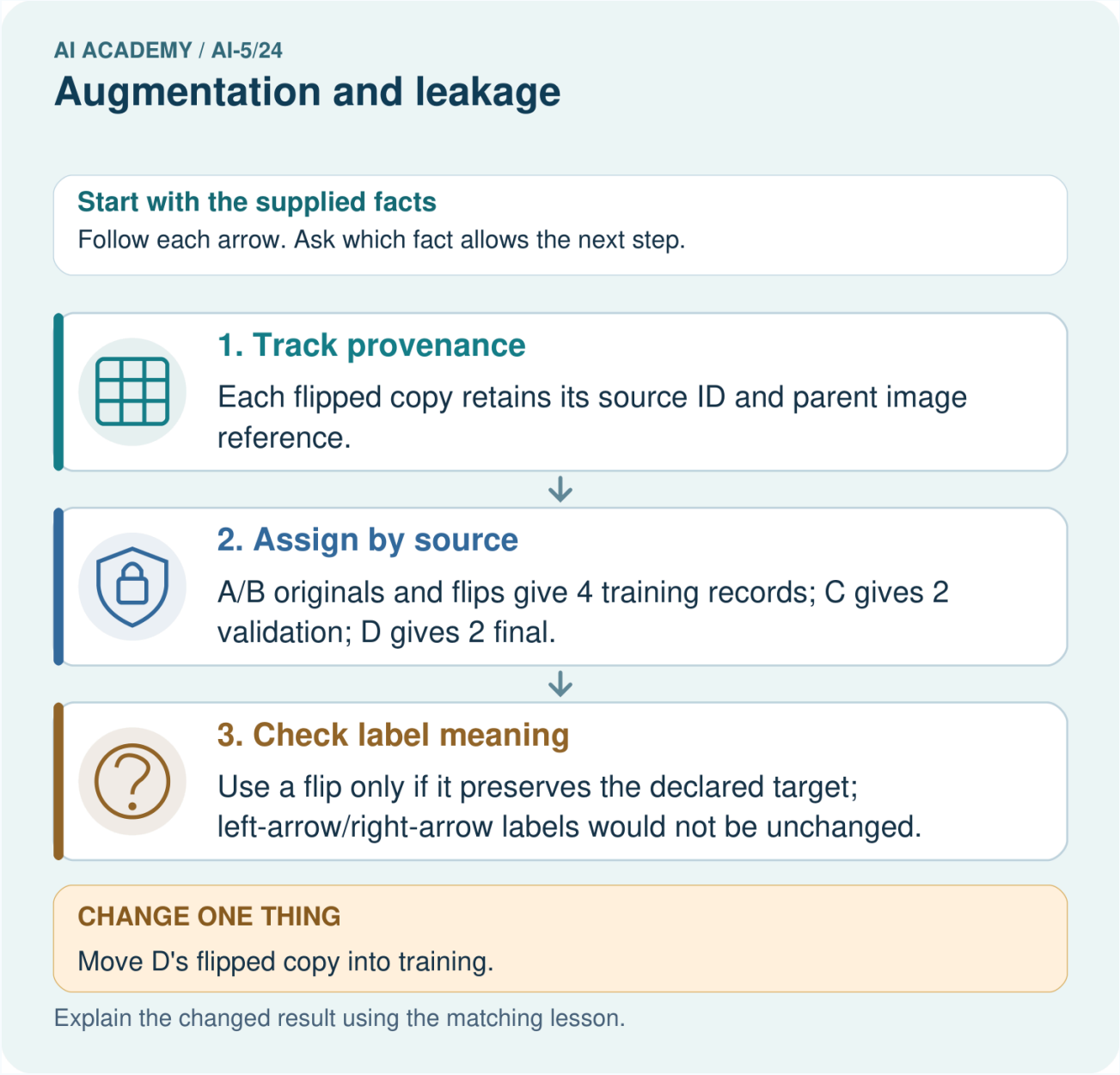
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Augmentation and leakage

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit lineage,duplicate pixels and the meaning of unseen sources

> Each audit catches a different failure;metadata alone does not verify real independence.

Complete previous-week relation and split-repair explanation

Week 23 used deliberately constructed toy images. Its validation E is exactly a row-swapped version of **training** A, and F is a row-swapped version of C.

They had different pixel arrays, but the transformation relation means they are a weak independence test for an experiment that trains on row-swapped variants.

If those variants are added, they become exact **training**-validation duplicates. Preserve the original narrow claim and design a new grouped split for the **augmentation** experiment.

Source groups should match the evaluation question. If the goal is generalisation to new objects or people, several photographs of the same object or person may need one group even when they have different original file IDs.

A lineage audit catches recorded relationships; an exact-pixel audit catches duplicates with inconsistent IDs. Neither audit alone proves that all dependence has been removed.

When a leak is discovered after validation has influenced choices, repair the split and rerun **training** and selection.

Do not merely rename or remove the overlapping files and keep claiming the old score evaluates the repaired experiment. Final data that influenced decisions is no longer reserved.

All audit conditions and justified actions

Complete development audit	Evidence and required action
Week23E/FversusA/C	E is the row-swapped Aarray; F is row-swapped C. Adding those variants to training creates exact training/validation duplicate pixels.
Recorded same source in two roles	Keep the whole source family in one split and re-establish suitable separate evaluation groups.

Complete development audit	Evidence and required action
Different claimed sources but same pixel array across splits	Audit exact duplicates as well as metadata; do not treat a fresh filename as unseen input. Pixel equality alone does not establish real-object identity.
Shared underlying object but different pixels/filenames	Source groups must match the evaluation question; recover object/session/parent lineage where required.
Validation already selected a model before repair	Preserve the old result's status and rerun affected fitting/selection under the corrected protocol. No repaired score is supplied here.

Try the P crop overlap and rerun decision

- **Training** contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

  Identify the overlap and explain what must be rerun if this validation set already selected a model.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise target and lineage reasoning with less help

> Write both the allowed transformation and retained source identity.

Current task and response

- For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

> **A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Augmentation** and **leakage**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Augmentation** and **leakage**. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured independent-source count

- How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent six-source count

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Keep the equality boundary

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete label,provenance and budget audit

> All original arrays,derived-record rules,faulty manifests and count assumptions are supplied.

Current tie task and original transform functions

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

```
def difference(x):
    return x[0][0] + x[1][0] - x[0][1] - x[1][1]

def label(x):
    return int(difference(x) > 0)

def swap_rows(x):
    return [x[1][:], x[0][:]]

def swap_columns(x):
    return [row[::-1] for row in x]

x = [[1, 0], [0, 0]]
assert label(x) == label(swap_rows(x)) == 1
assert label(swap_columns(x)) == 0
tie = [[1, 0], [0, 1]]
assert label(tie) == label(swap_columns(tie)) == 0
assert x == [[1, 0], [0, 0]]
```

Complete P–T source registry and retention counts

Complete fictional original source · Binary image rows · Class under known rule · Assigned role · Retention policy

> **Complete fictional original source:** P
>
> **Binary image rows:** [1, 0]; [0, 0]

Class under known rule: 1

Assigned role: Training

Retention policy: Original and row swap

Complete fictional original source: Q

Binary image rows: [0, 1]; [0, 0]

Class under known rule: 0

Assigned role: Training

Retention policy: Original and row swap

Complete fictional original source: R

Binary image rows: [1, 0]; [1, 0]

Class under known rule: 1

Assigned role: Validation

Retention policy: Original only

Complete fictional original source: S

Binary image rows: [0, 1]; [0, 1]

Class under known rule: 0

Assigned role: Final

Retention policy: Original only

Complete fictional original source: T

Binary image rows: [1, 1]; [0, 0]

Class under known rule: 0

Assigned role: Final

Retention policy: Original only

Registry result	Source groups	Retained records	Derived evaluation records excluded
Training P/Q	2	4	0
Validation R	1	1	1
Final S/T	2	2	2
Total	5	7	3

Complete runnable label and overlap audit

```
# Complete fictional provenance audit. No model is trained and no new score is claimed.
from copy import deepcopy
from itertools import product

def difference(x):
    return x[0][0]+x[1][0]-x[0][1]-x[1][1]
def label(x):
    return int(difference(x)>0)
def swap_rows(x):
    return [x[1][:],x[0][:]]
def swap_columns(x):
    return [row[::-1] for row in x]

# Every binary 2x2 pattern checks the known mathematical rule, not unknown real labels.
for bits in product([0,1],repeat=4):
    x=[list(bits[:2]),list(bits[2:])]
    assert label(swap_rows(x))==label(x)
    assert difference(swap_columns(x))==-difference(x)
    expected=0 if difference(x)==0 else 1-label(x)
    assert label(swap_columns(x))==expected
print('all 16 binary patterns: row swaps preserve labels; column swaps require the
declared rule')

sources = {
    'P': ([[1,0],[0,0]], 'training'),
    'Q': ([[0,1],[0,0]], 'training'),
    'R': ([[1,0],[1,0]], 'validation'),
    'S': ([[0,1],[0,1]], 'final'),
    'T': ([[1,1],[0,0]], 'final'),
}
```

```
inventory=[]
for group,(image,split) in sources.items():
    for transform,pixels in [('original',deepcopy(image)),('row_swap',swap_rows(image))]:
        inventory.append({'id':group+'-'+transform,'source':group,'parent':None if
transform=='original' else group+'-original',

'split':split,'transform':transform,'pixels':pixels,'label':label(pixels)})
retained=[r for r in inventory if r['split']=='training' or r['transform']=='original']

def audit(records):
    groups={};pixels={}
    for r in records:
        groups.setdefault(r['source'],set()).add(r['split'])
        key=tuple(tuple(row) for row in r['pixels'])
        pixels.setdefault(key,[]).append((r['id'],r['split']))
    shared_groups={g:sorted(s) for g,s in groups.items() if len(s)>1}
    duplicate_pixels=[pairs for pairs in pixels.values() if len({split for _,split in
pairs})>1]
    return shared_groups,duplicate_pixels

for split in ['training','validation','final']:
    part=[r for r in retained if r['split']==split]
    print(split,'groups / retained records',len({r['source'] for r in part}),len(part))
print('retained / excluded',len(retained),len(inventory)-len(retained))
print('clean recorded-group / exact-pixel overlaps',audit(retained))

bad_group=deepcopy(retained)
next(r for r in bad_group if r['id']=='P-row_swap')['split']='validation'
print('new filename does not repair shared source:',audit(bad_group)[0])

alias=deepcopy(next(r for r in retained if r['id']=='P-row_swap'))
alias.update(id='claimed-U-original',source='claimed-
U',parent=None,split='validation',transform='original')
print('different claimed source, same pixels:',audit(retained+[alias]))
print('Audit only: exact pixels do not prove real-object identity; missing provenance
still needs review.')
print('Any affected fitting/selection must be rerun after a justified split repair; no
repaired score is supplied.')
```

Full separate training-budget comparison

Separate budget comparison; batch size 1 · Records per epoch · Epochs · Updates · Source identities

> **Separate budget comparison; batch size 1:** 12 originals, eachwith 2 variants
>
> **Records per epoch:** 36
>
> **Epochs:** 5
>
> **Updates:** 180
>
> **Source identities:** 12

> **Separate budget comparison; batch size 1:** 12 originals only
>
> **Records per epoch:** 12
>
> **Epochs:** 5
>
> **Updates:** 60
>
> **Source identities:** 12

> **Separate budget comparison; batch size 1:** One declared equal-update baseline option
>
> **Records per epoch:** 12
>
> **Epochs:** 15
>
> **Updates:** 180
>
> **Source identities:** 12

> **Separate budget comparison; batch size 1:** Independent source count meaning
>
> **Records per epoch:** Variants are related to their originals
>
> **Epochs:** Record exposure policy
>
> **Updates:** Matched updates do not prove benefit
>
> **Source identities:** 36 files are not 36 independent subjects

Normal,alias,missing-provenance and budget probes

Complete audit probe	Expected result / action
All16 binary patterns under known rule	Row swaps preserve classes; column swaps negate d; ties stay 0
Clean P–T retained manifest	Training2 groups/4 records, validation 1/1, final 2/2; 7 retained/3 derived excluded
Move P-row_swap to validation	Recorded source P now crosses training/validation despite its different filename
Alias P-row_swap as claimed-U-original in validation	No common claimed source, but exact pixels duplicate a training row; investigate provenance
Missing parent/group or ambiguous transformed label	Review before splitting/retaining; do not invent independence or a new real-world label
Eight originals plus 2 variants, 4 epochs, batch 1	24 records×4=96 updates versus 8×4=32; predeclare a matched budget
Any repaired split or augmentation plan	No observed new model score supplied; rerun and report actual results rather than declare improvement

Predict,derive,audit and record unmeasured effects

1. Predict labels and all registry counts before running or tracing the complete audit program. It supplies every P–T original, creates row-swap records with source/parent metadata, and compares a clean plan with two explicit faulty manifests. The exact-pixel audit checks unseen input overlap; it cannot infer real-object identity or correct missing provenance.

   The registry is a public paper exercise, not a secret scored dataset or a new training run. Separately compare the complete update budgets with matched starts, architecture, evaluation cases and declared exposure policy. Report unknown effects honestly:there is no measured augmentation benefit here.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: count retained records in a new grouped split

> Apply the supplied training/evaluation retention rules without mixing groups.

Complete independent six-source assignment

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a grouped split independently

Attempt this without hints

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep transformations,lineage and budgets in the project record

> A reader should know which labels,source families and training exposures changed.

Project artifact and evidence table

Add a labelled **augmentation** and **leakage evidence** sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a justified transformation and a source-aware count

> Explain why more files and more updates are different from more independent evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Check whether a transformation preserves the task label

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Augmentation	Creating transformed training examples under a declared procedure
Label preservation	A transformation leaving the correct target unchanged for the stated task
Source group	An identifier joining records that must stay together when splitting
Leakage	Information crossing an evaluation boundary in a way that undermines the intended independent check

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

- **Training** contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

  Identify the overlap and explain what must be rerun if this validation set already selected a model.

- Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

  Give both update counts and one way to match budgets without using final outcomes.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent J–Ogroup assignment and retained-record count for the first individual attempt.

Current explanation and response

- Suppose 12 **training** source images each produce an original plus two variants. That creates 36 records but still only 12 **source group**s. Keep validation and final groups separate.

  Report source counts as well as record counts; repeated variants are correlated **observation**s, not new independent subjects.

- With one-example batches the runs use 180 and 60 updates. A declared comparison could give both 180 updates, cycle the baseline data and record the exposure policy.

  Keep starting parameters, architecture and evaluation cases matched where possible. Select **augmentation** settings using validation **evidence** and report the limits of the chosen budget.

  Next week moves from images to sequence representations.

- Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

  Give both update counts and one way to match budgets without using final outcomes.

An augmented run uses 36 records for five epochs while a baseline uses 12 records for five epochs. Both use one-example batches. Are their update budgets matched? Propose a controlled comparison and explain what 36 records does not establish.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Keep every augmented image with its original **source group**.

By the end, I can	How I will show it
Check whether a transformation preserves the task label	Explain the deciding step, show a trace or test, and state one limit.
Keep related examples in one split using source groups	Explain the deciding step, show a trace or test, and state one limit.
Compare augmentation fairly without inflating independent evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Keep every augmented image with its original **source group**.
- Which labelled step matches this detail? Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.
- 1. Assign by source
- 2. Track provenance
- 3. Check label meaning
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check label meaning Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Changed-case question and optional hint

- Changed-case practice: Move D's flipped copy into **training**. Which conclusion is supported by the supplied facts?
- 1. Final-test group separation fails.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Repair and retain the record

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

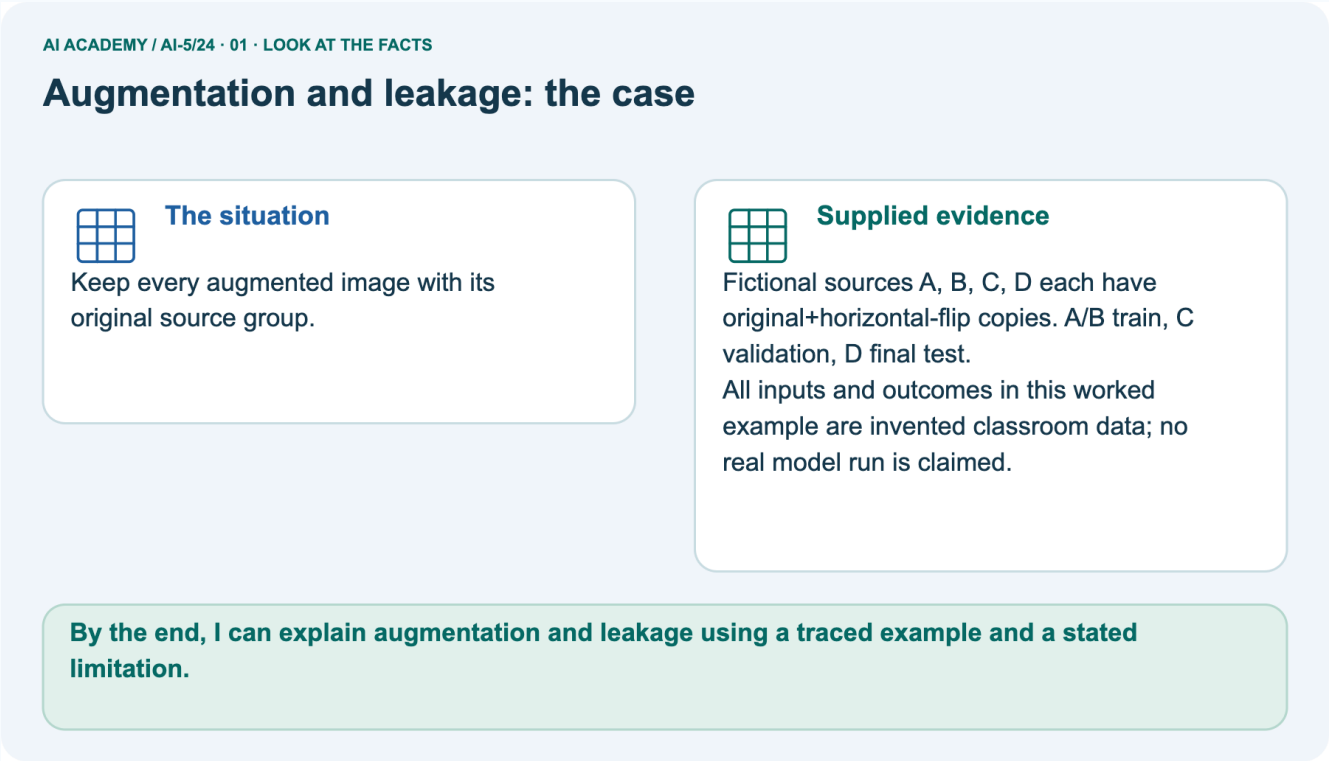
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Augmentation and leakage: the case



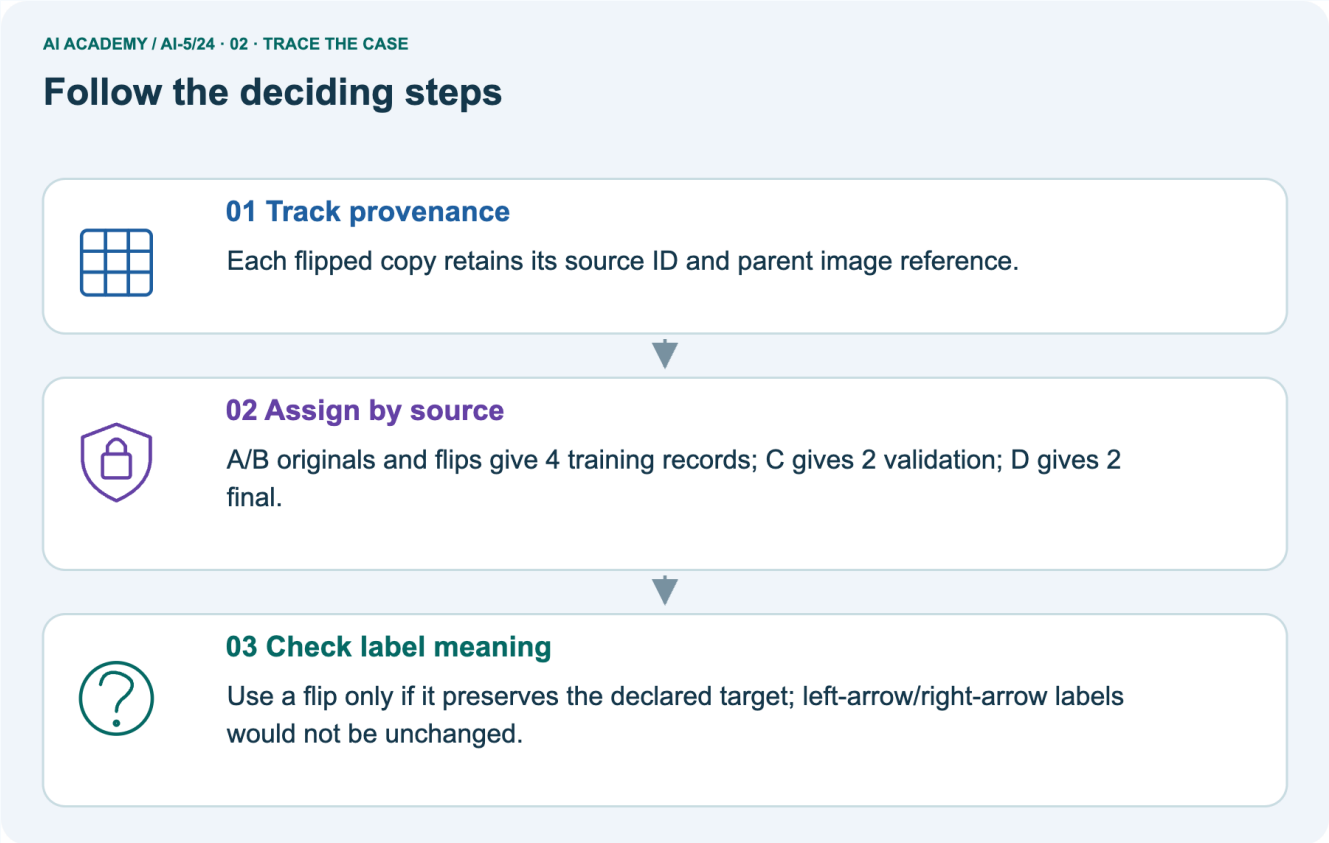
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep every augmented image with its original **source group**. Supplied facts: Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Track provenance: Each flipped copy retains its source ID and parent image reference. → Assign by source: A/B originals and flips give 4 **training** records; C gives 2 validation; D gives 2 final. → Check label meaning: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Point to each numbered stage and explain which recorded **input** it uses.

03. Every variant stays with its source group

AI ACADEMY / AI-5/24 · 03 · INSPECT THE LESSON MODEL

Every variant stays with its source group

Source group	Assigned split	Derived records
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

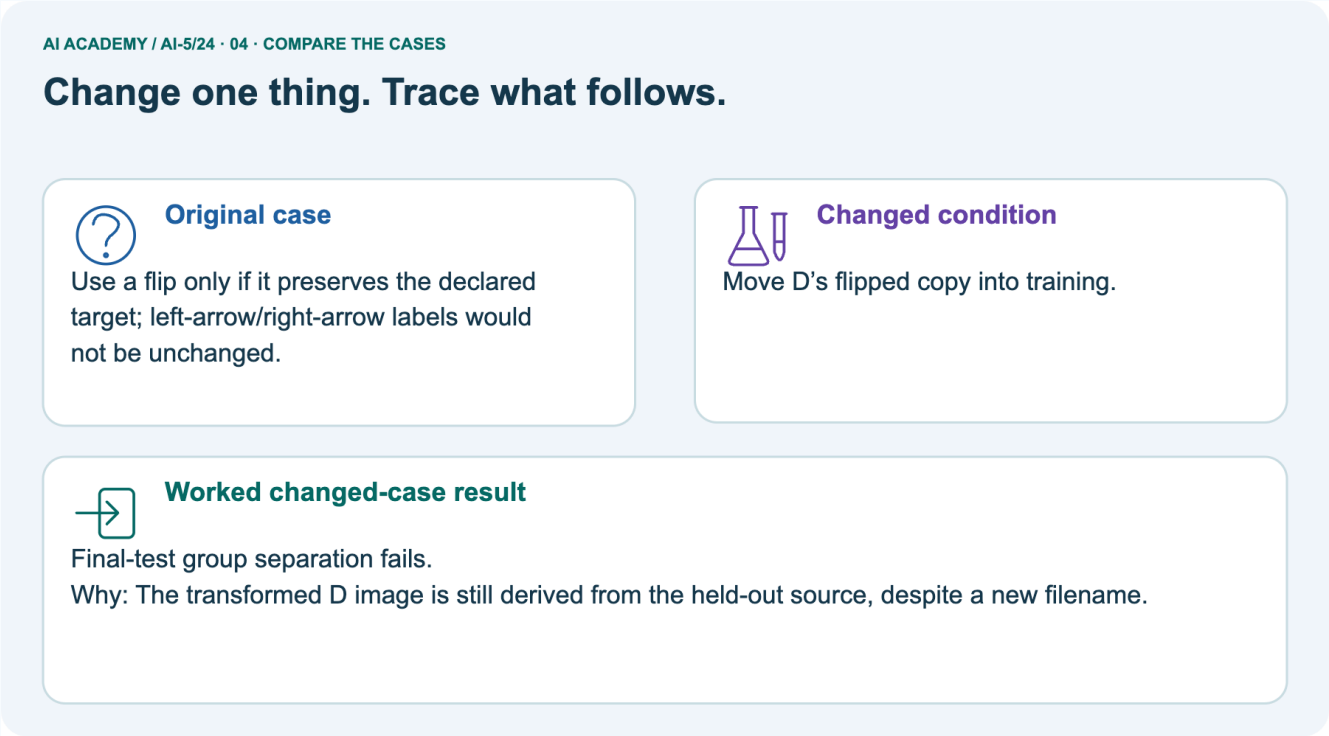
Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Every variant stays with its **source group**. A | **Training** | A original and A row swap. B | Validation | B original only. C | Final test | C original only

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



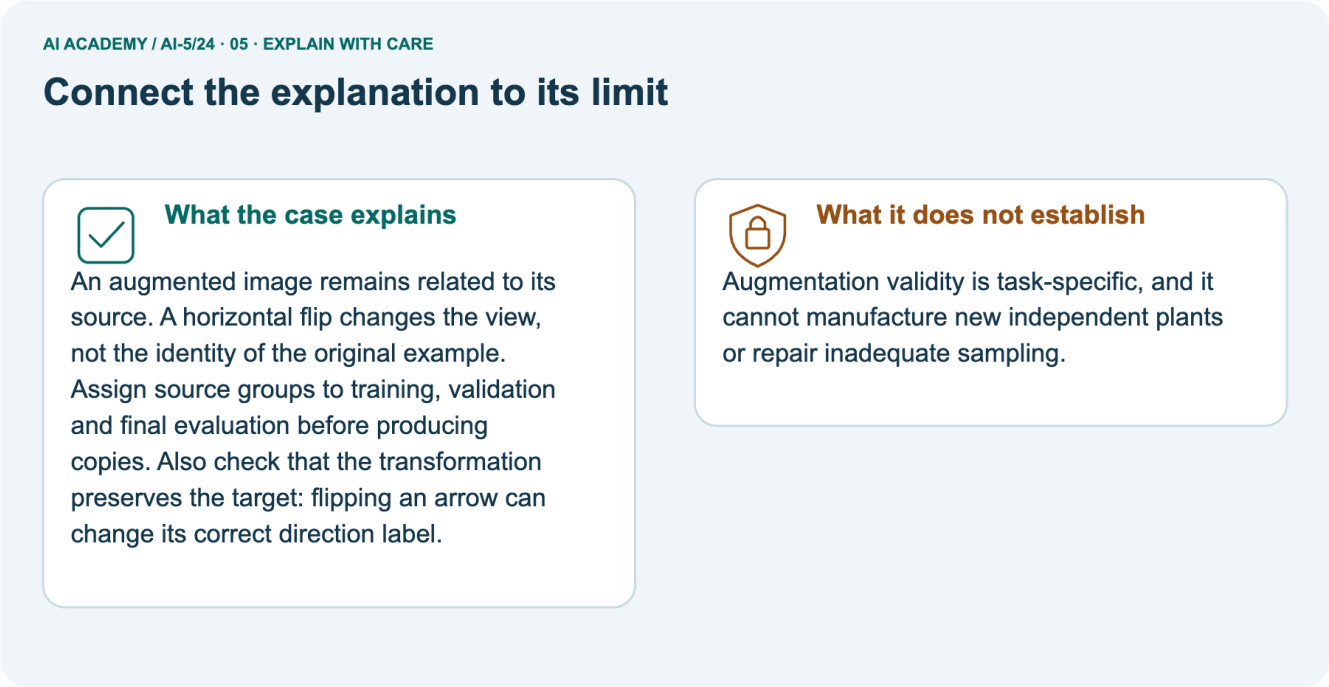
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. Changed condition: Move D's flipped copy into **training**.

Expected result: Final-test group separation fails. Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

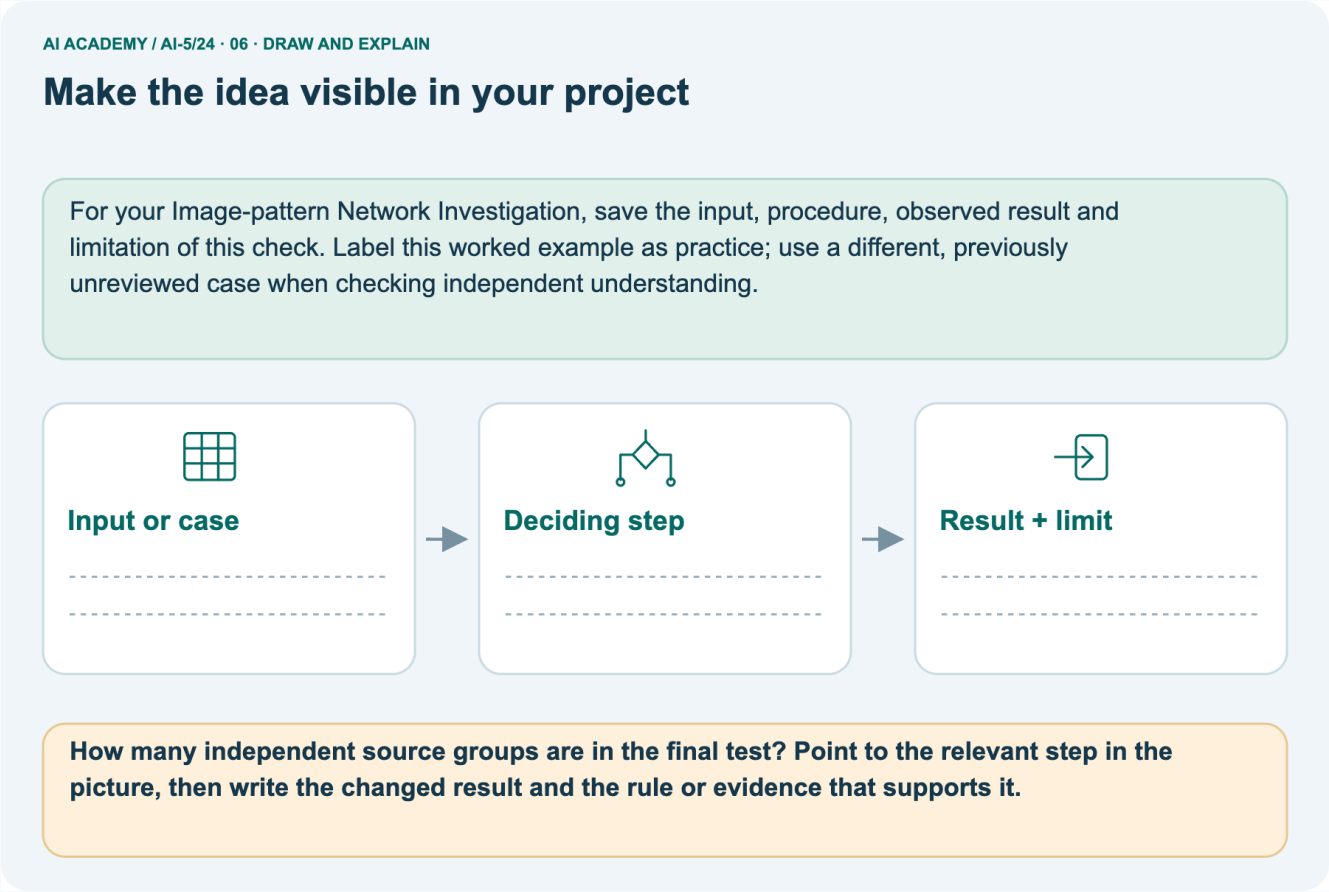
An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Limit: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Augmentation and leakage: the case

02. Follow the deciding steps

03. Every variant stays with its source group

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.